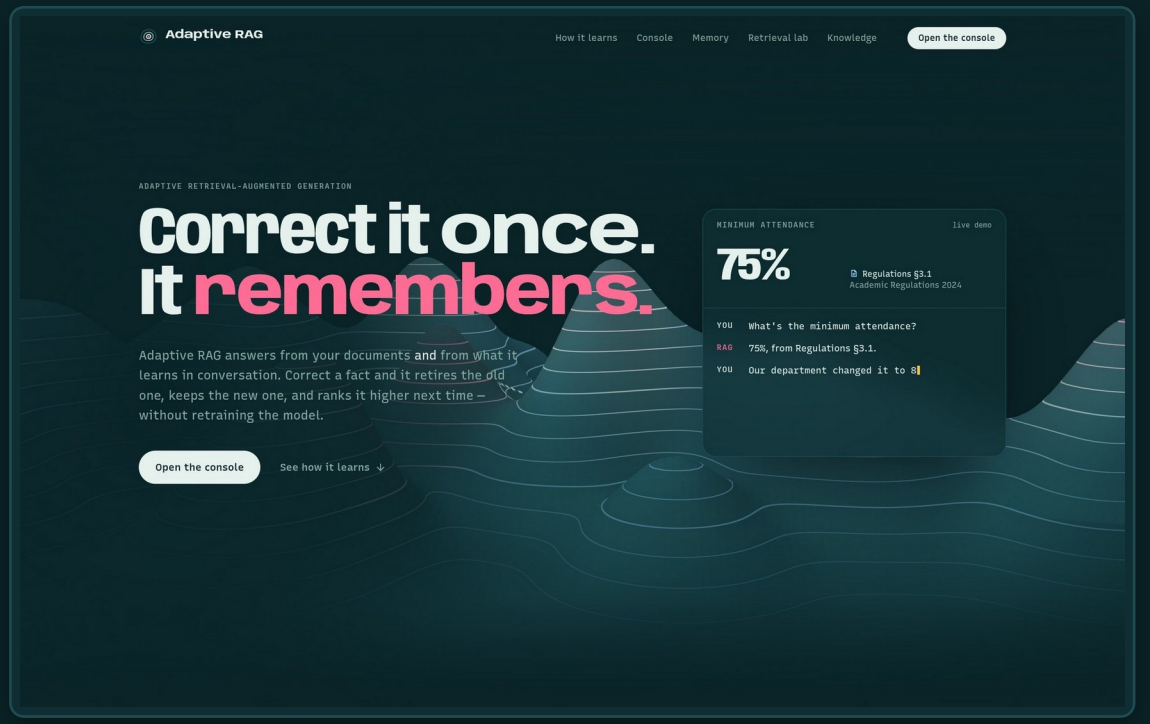
AI COURSE PROJECT

Adaptive RAG

Correct it once. It remembers.

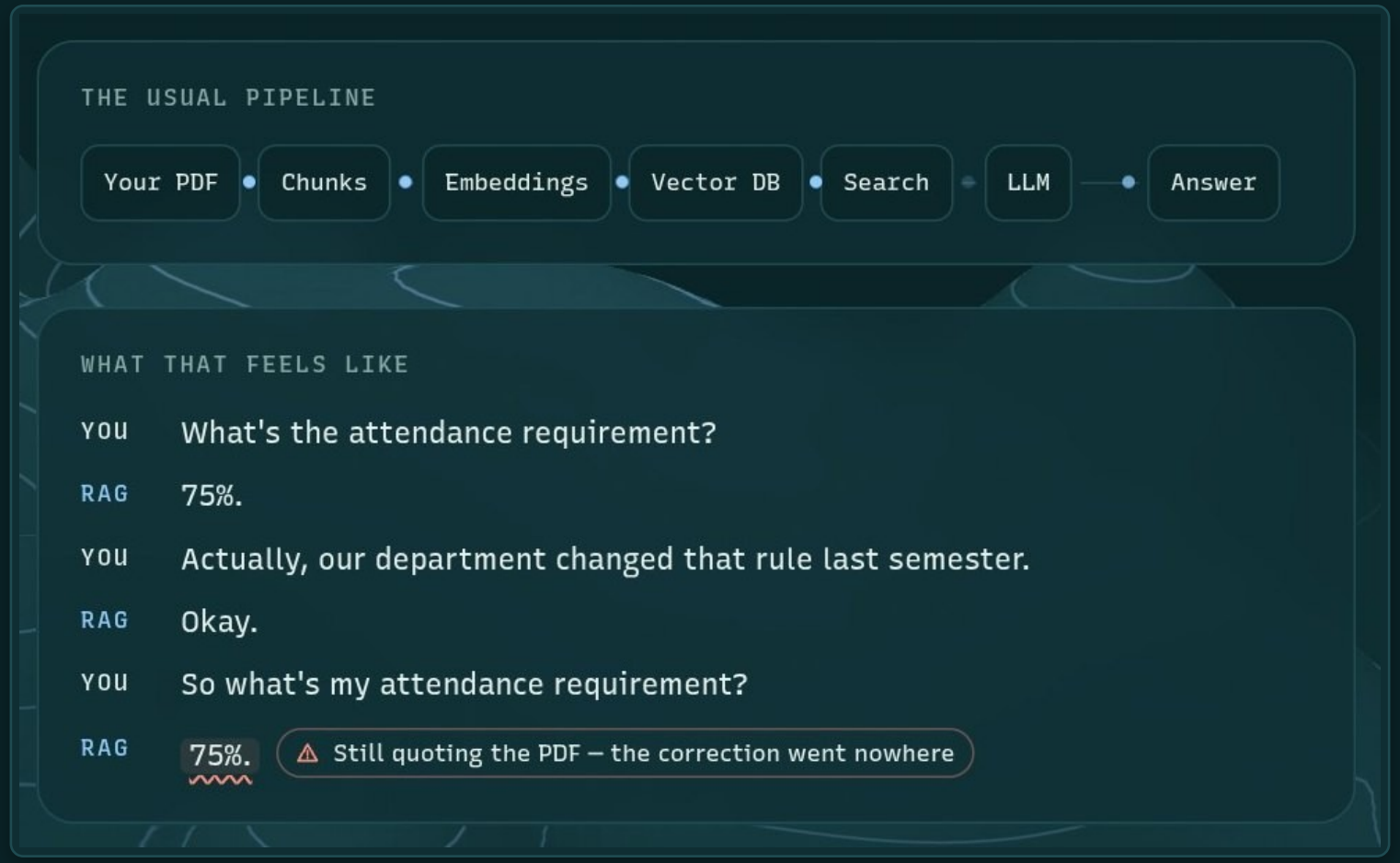
A study assistant that learns from what you tell it — corrections, preferences, short-lived facts — and re-ranks what it retrieves, without retraining the language model.

A tour of every feature of the website, with screenshots



THE PROBLEM

Ordinary RAG answers from a frozen page



From the landing page: the student corrects the rule, and the next answer still quotes 75%.

⚠ Three gaps in a plain RAG pipeline

Corrections go nowhere

Tell it the rule changed to 80% and the next answer still quotes 75% from the PDF.

Every source is trusted equally

A wrong or outdated passage keeps coming back, forever.

Ranked by similarity only

The passage that sounds closest wins, even when it's the wrong source for the question.

THE IDEA

Learn around the model, not inside it



LEARNS

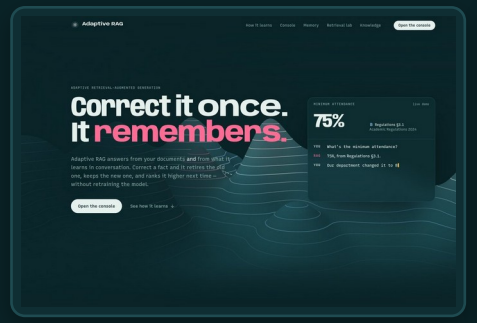
- Memories — added, replaced, expired
- How much each source is trusted
- How the four ranking signals are weighed

STAYS FIXED

- The language model — never trained or fine-tuned
- Your original documents
- Everything is visible and can be undone

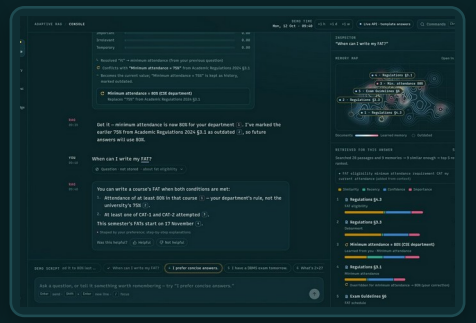
THE WEBSITE

Five pages, one system



Landing

/ · The story in one scroll over a live 3-D terrain



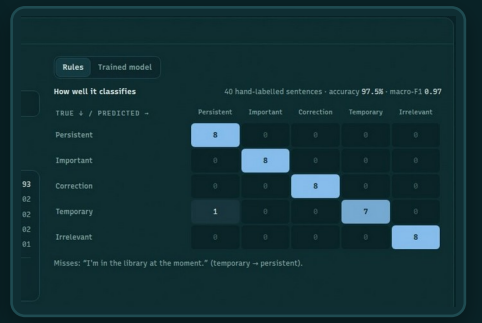
Console

/console · Chat with every pipeline step shown as it happens



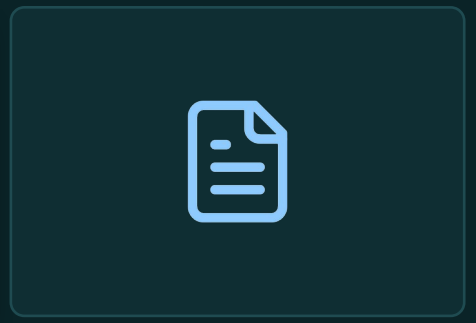
Memory

/memory · What it learned: cards, 3-D terrain, timeline



Retrieval lab

/retrieval · Ranking formula, learned weights, classifier accuracy



Knowledge

/knowledge · Documents, passages, and adding new files

ON EVERY APP PAGE

Demo clock
+1 h / +1 d / +1 w time travel

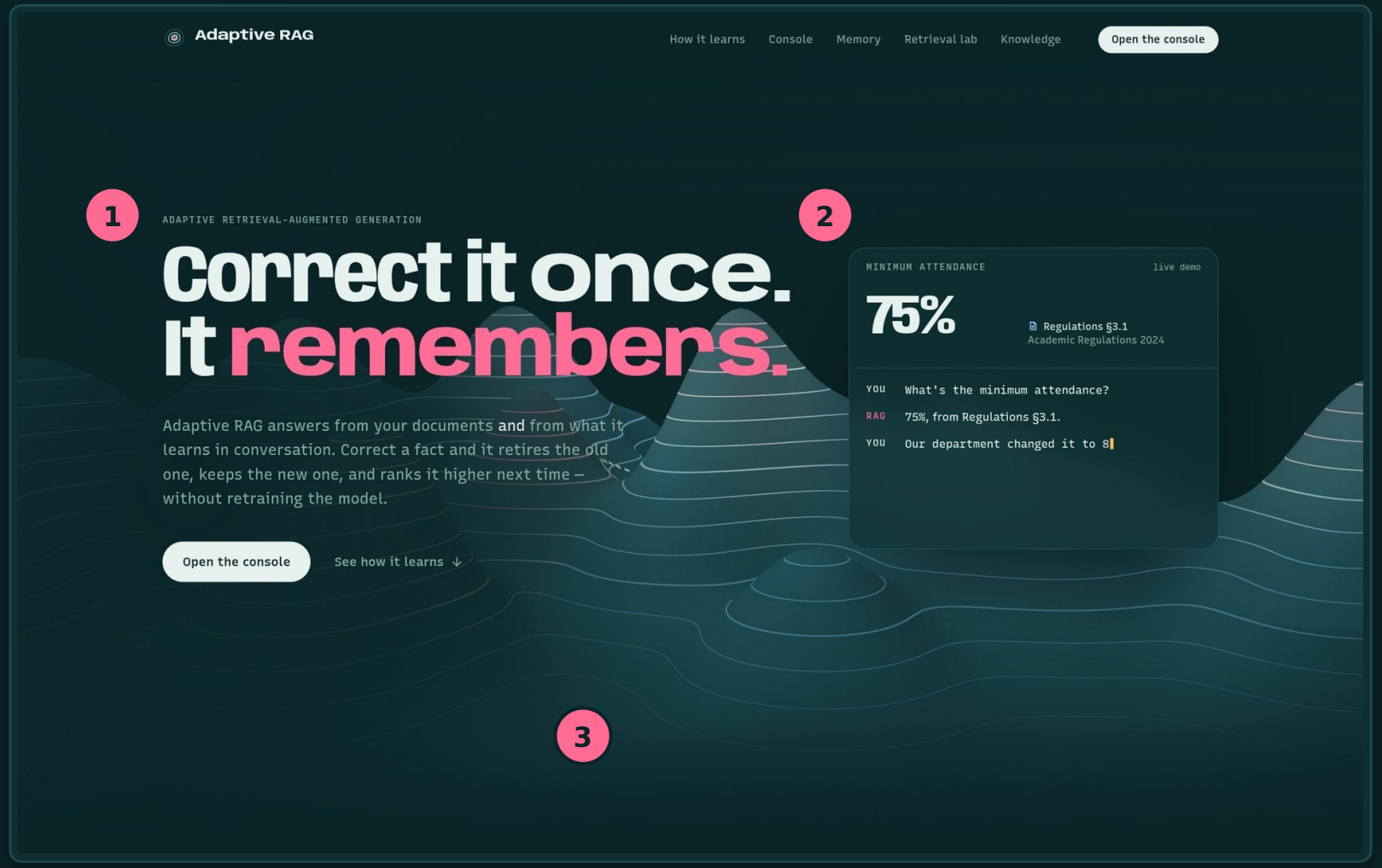
Command palette
Ctrl K — go anywhere, search memories

Toasts
Every memory change is announced

Live API badge
Shows which model answers

LANDING PAGE

The story in one scroll



1. **Headline + call to action**
   Opens the console or scrolls to how it learns.

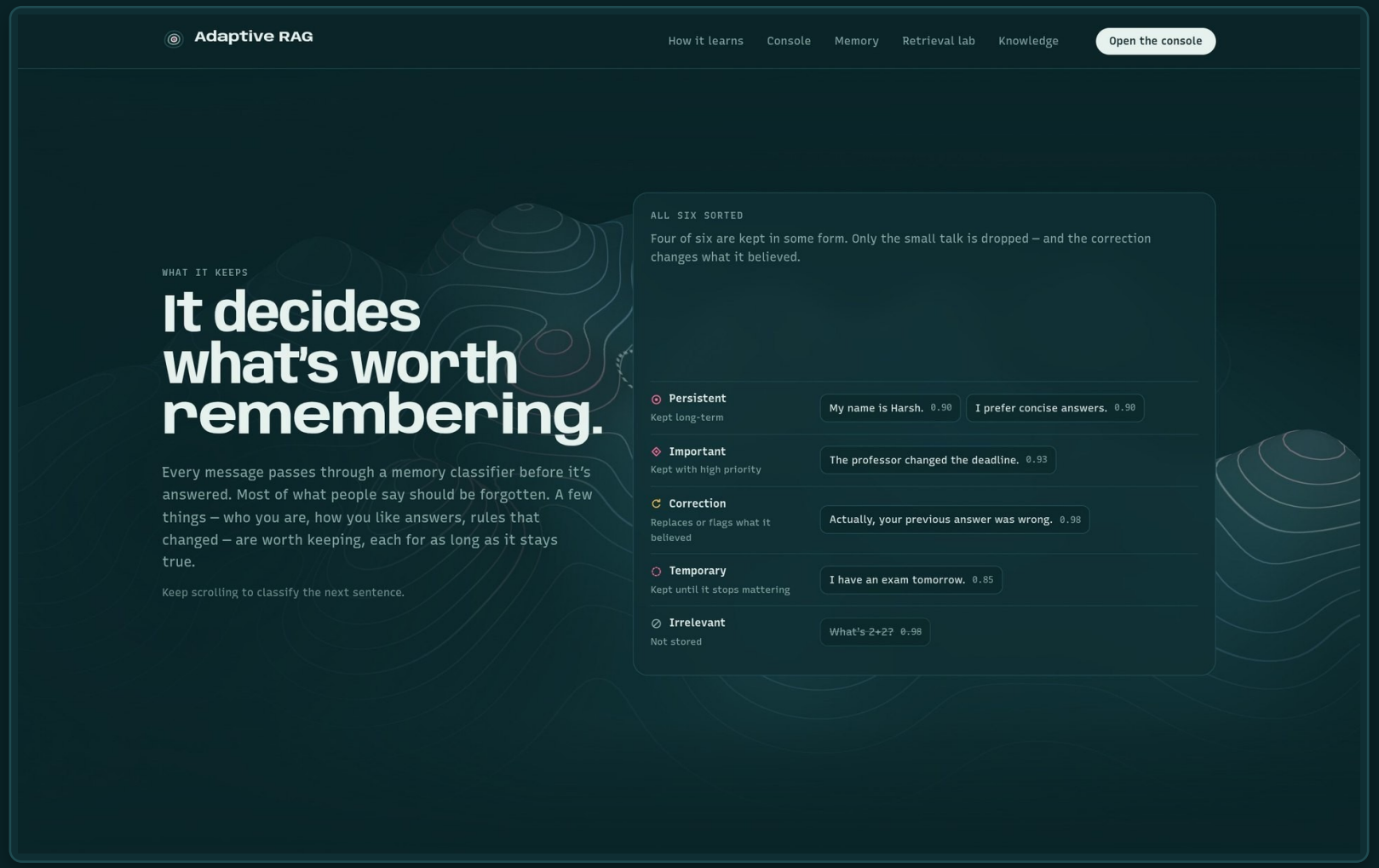
2. **Live correction demo**
   Auto-plays: asks, gets 75%, corrects to 80% — the card updates.

3. **3-D knowledge terrain**
   Fixed behind the page; each section moves the camera and reshapes the map.

EIGHT SECTIONS

Hero · The problem · What it keeps · Corrections · Ranking · The learning loop · Architecture · Try it

It decides what's worth remembering

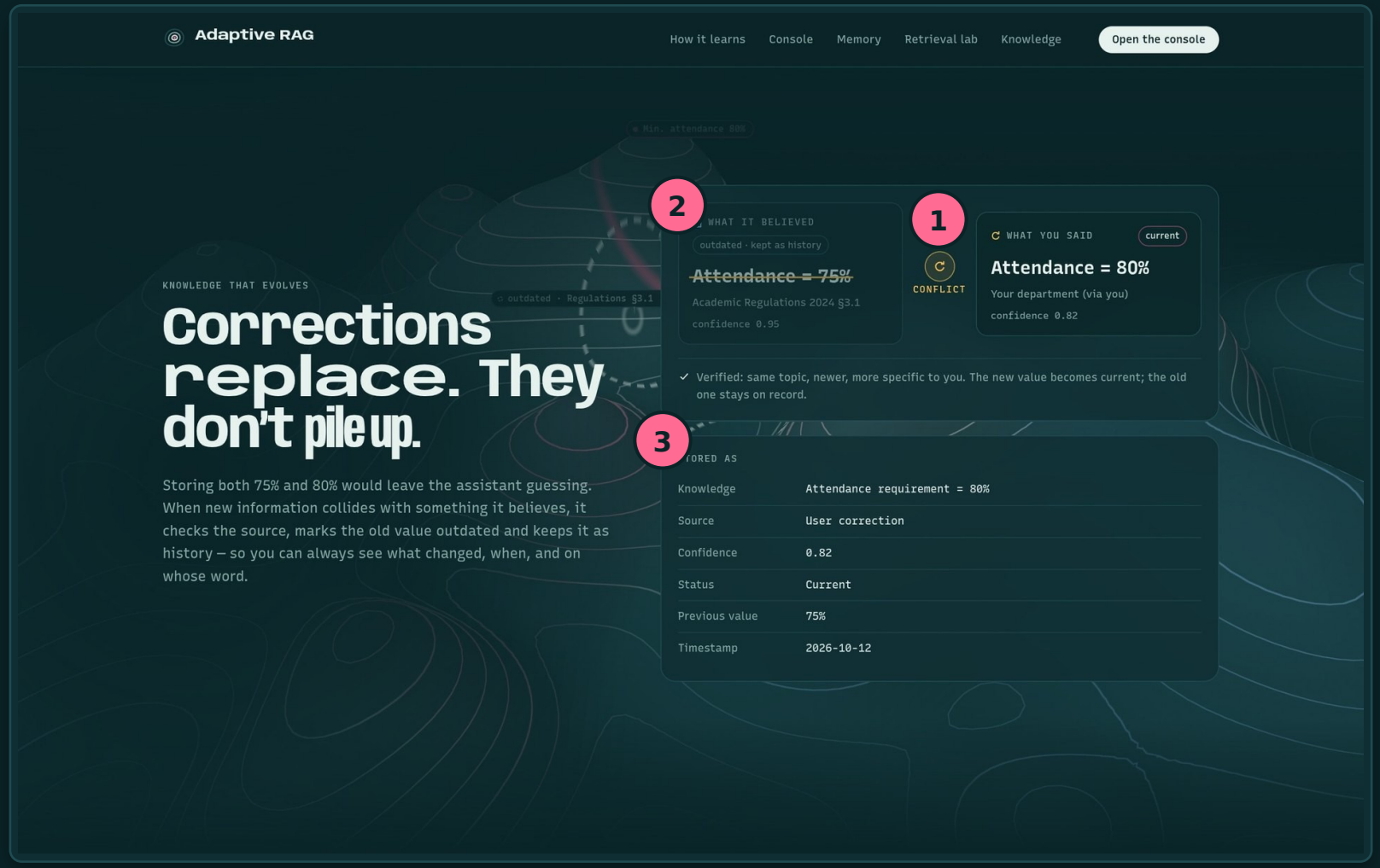


FIVE CLASSES

- **Persistent**
  About you — name, preferences
- **Important**
  New knowledge — a rule, a deadline
- **Correction**
  Replaces something it believed
- **Temporary**
  True for a while — kept until it expires
- **Irrelevant**
  Questions, small talk — not stored

Scroll-driven: each scroll step sorts the next sentence.

Corrections replace — they don't pile up

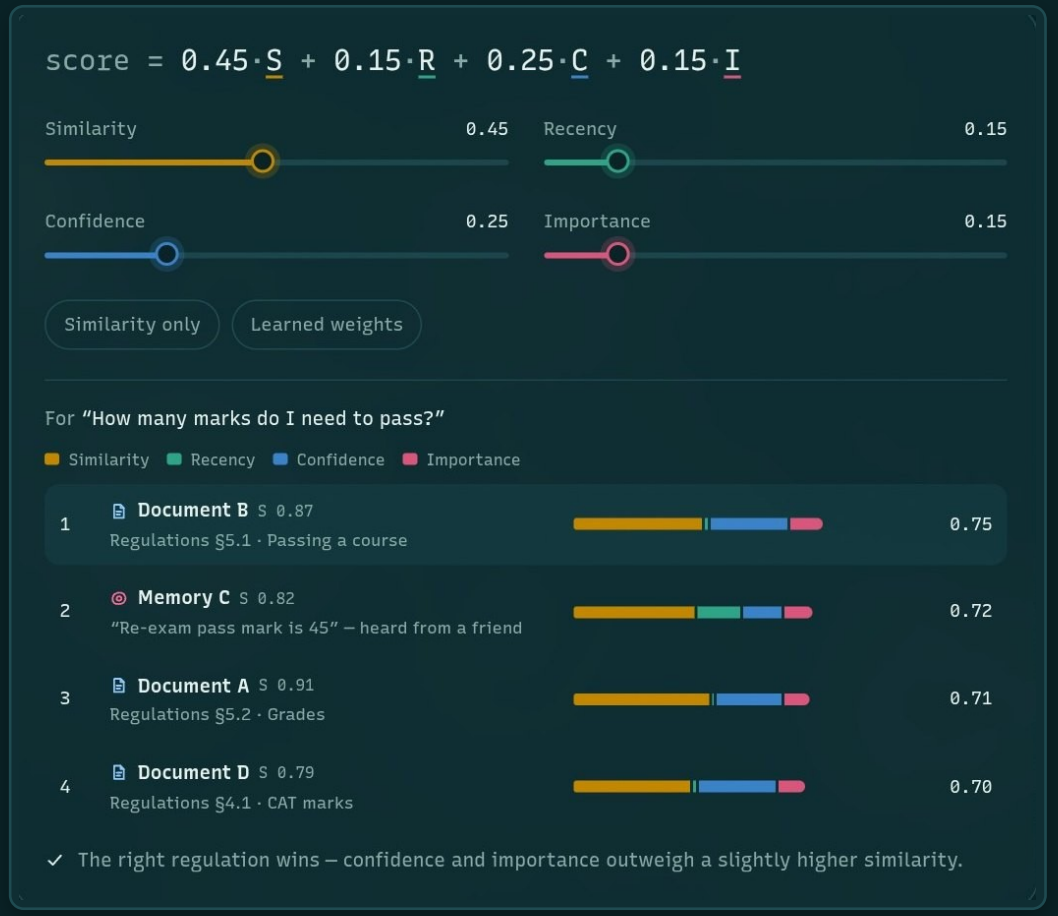


1 Conflict detected
Same topic (minimum attendance), different value: 75% vs 80%.

2 Old value kept as history
75% is marked outdated, not deleted — you see what changed.

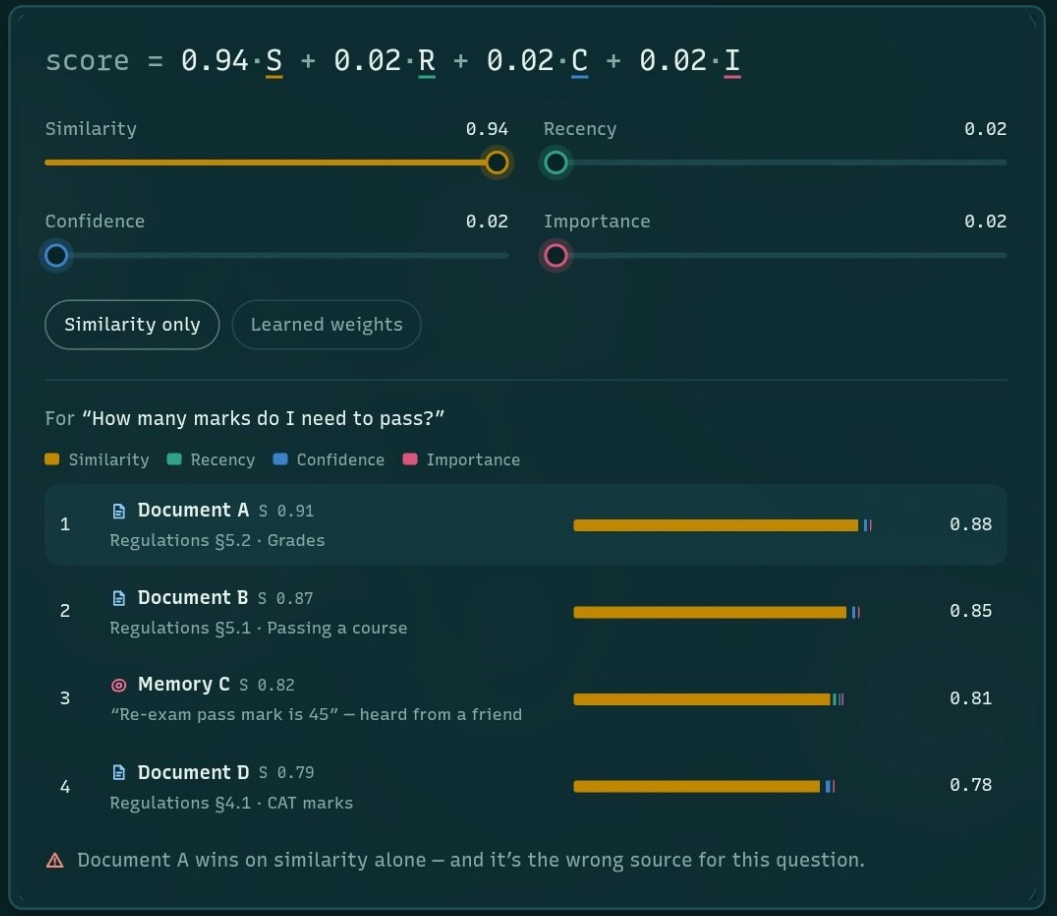
3 New value becomes current
Stored with its source (you), a confidence and the previous value.

Ranked by more than similarity



Learned weights

Right regulation wins: confidence and importance beat raw similarity.



Similarity only

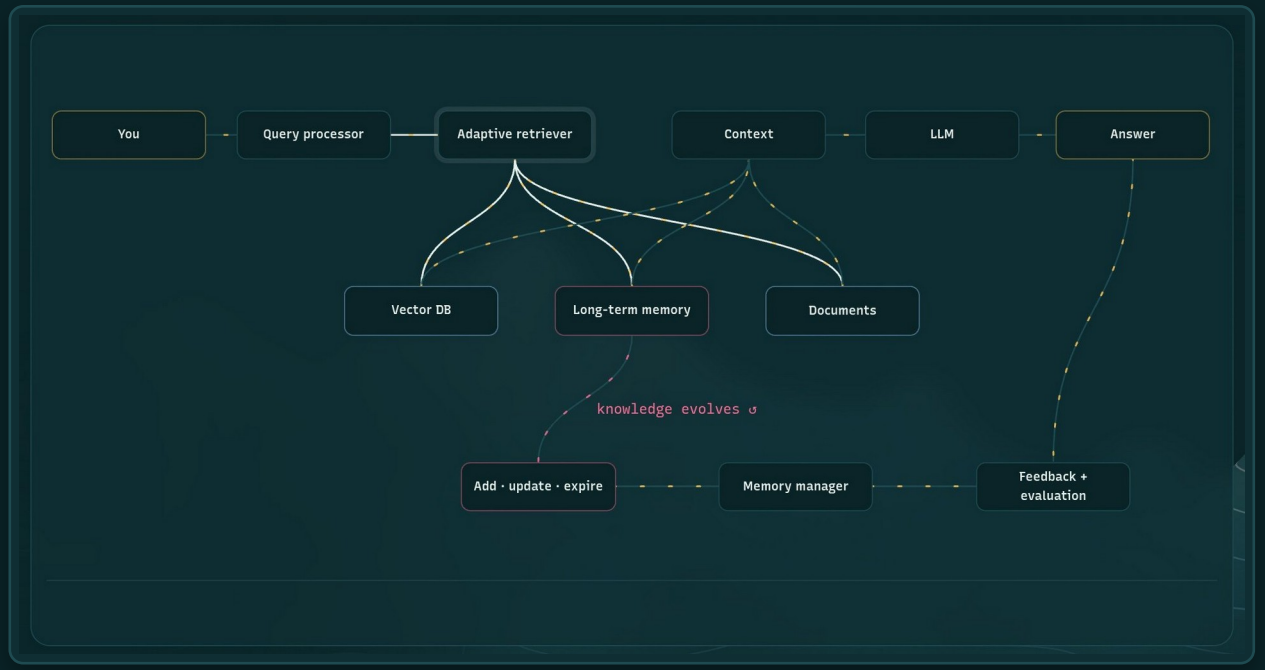
Closest-sounding text wins, even from the wrong source.

The model stays the same; everything around it learns



THE LEARNING LOOP

Question → retrieve → answer → your feedback → evaluate retrieval → update memory & weights. The language model in the centre is unchanged.

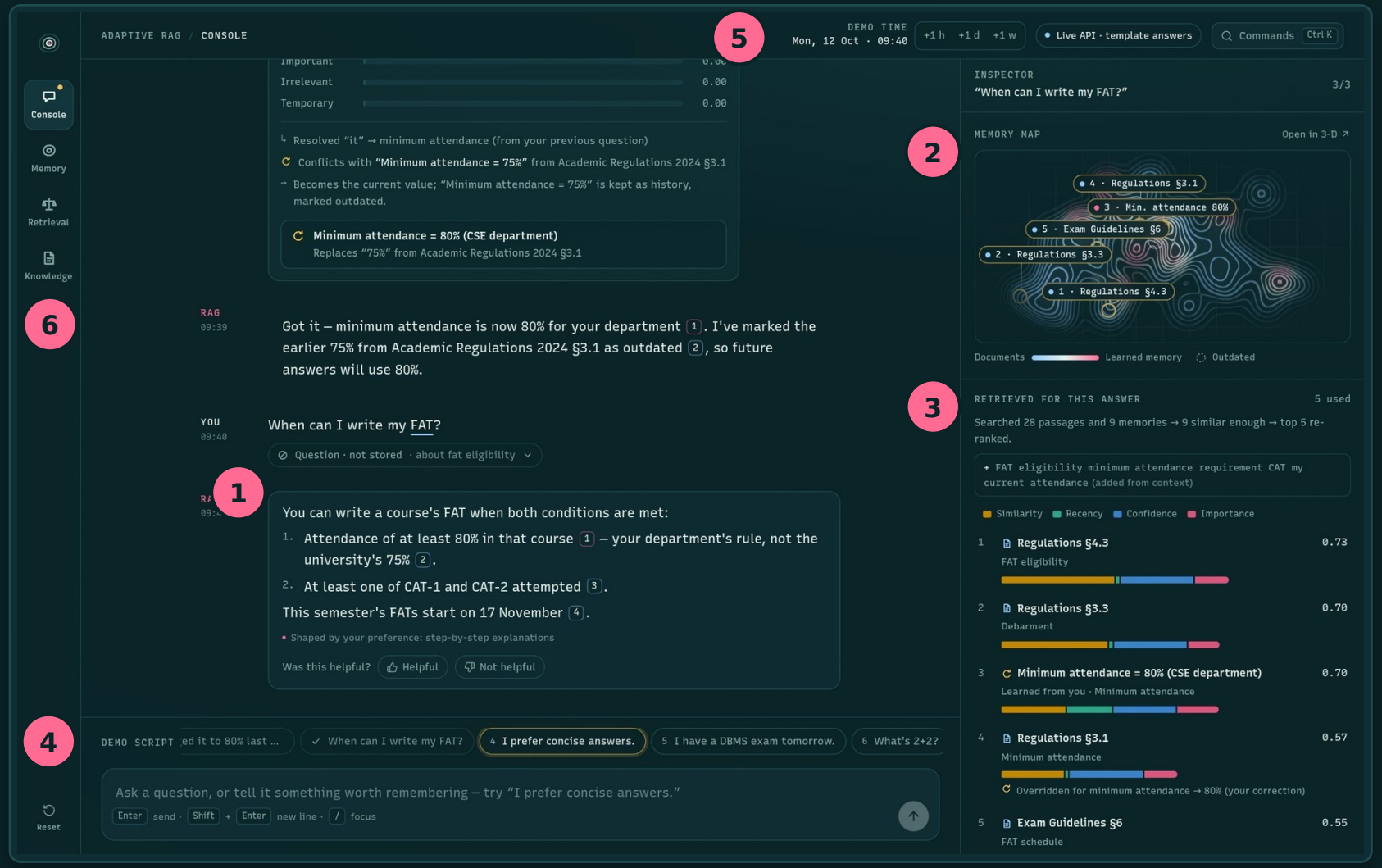


THE ARCHITECTURE, INTERACTIVE

Hover any block: the adaptive retriever reads the vector DB, long-term memory and documents, then hands context to the LLM.

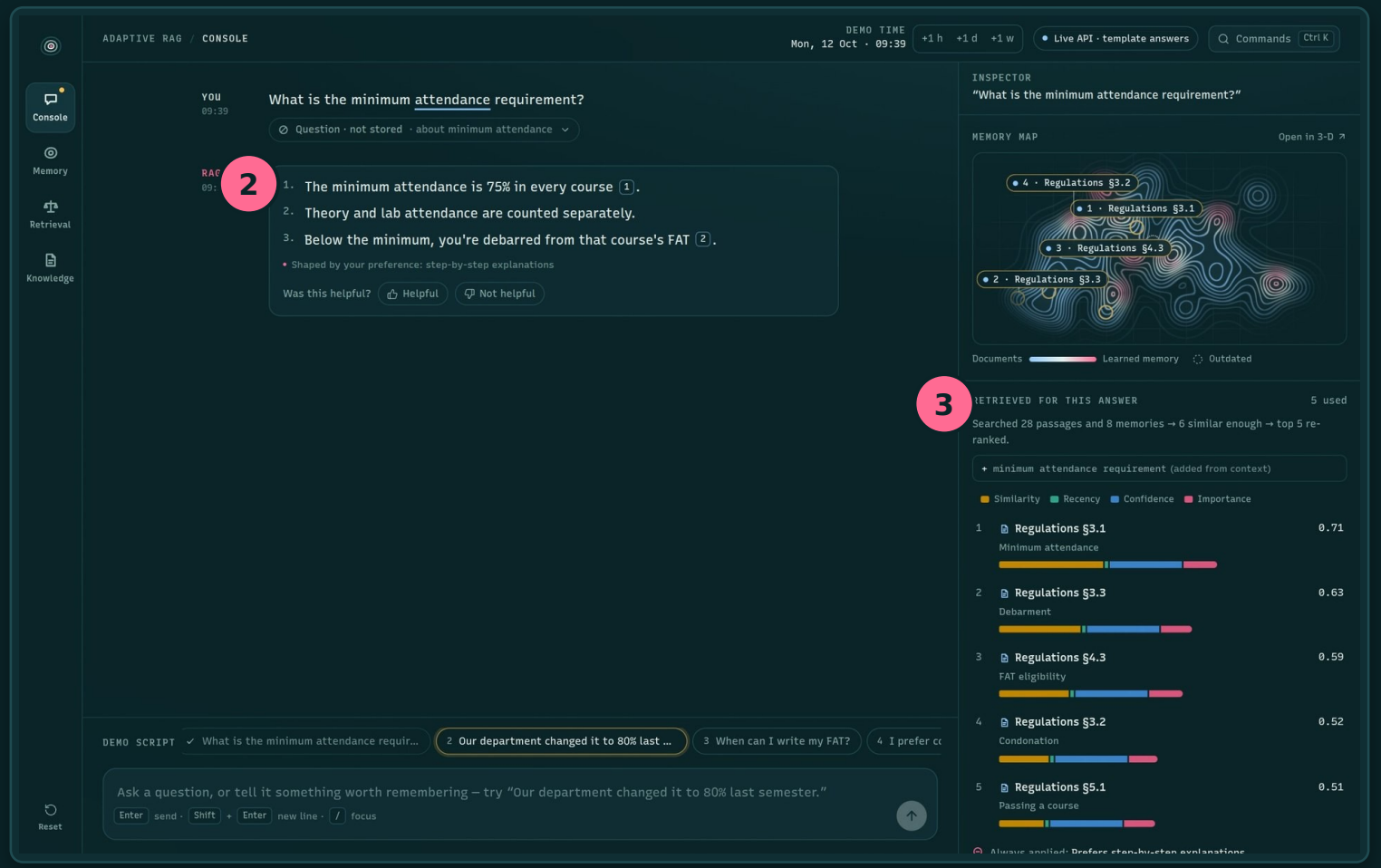
CONSOLE

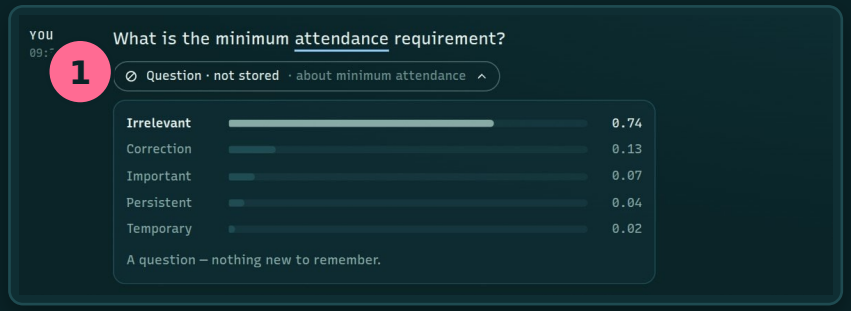
Chat with the whole pipeline on show



1. **Conversation**
   Classifier verdict, memory writes, cited answer.
2. **Memory map**
   Where the question landed; rings mark sources used.
3. **Retrieved for this answer**
   Top 5 sources with score bars (S, R, C, I).
4. **Demo script**
   Nine chips run the guided story.
5. **Demo clock + Live API**
   Time travel; which model answers.
6. **Navigation rail**
   Console · Memory · Retrieval · Knowledge.

Ask: it answers from the regulations





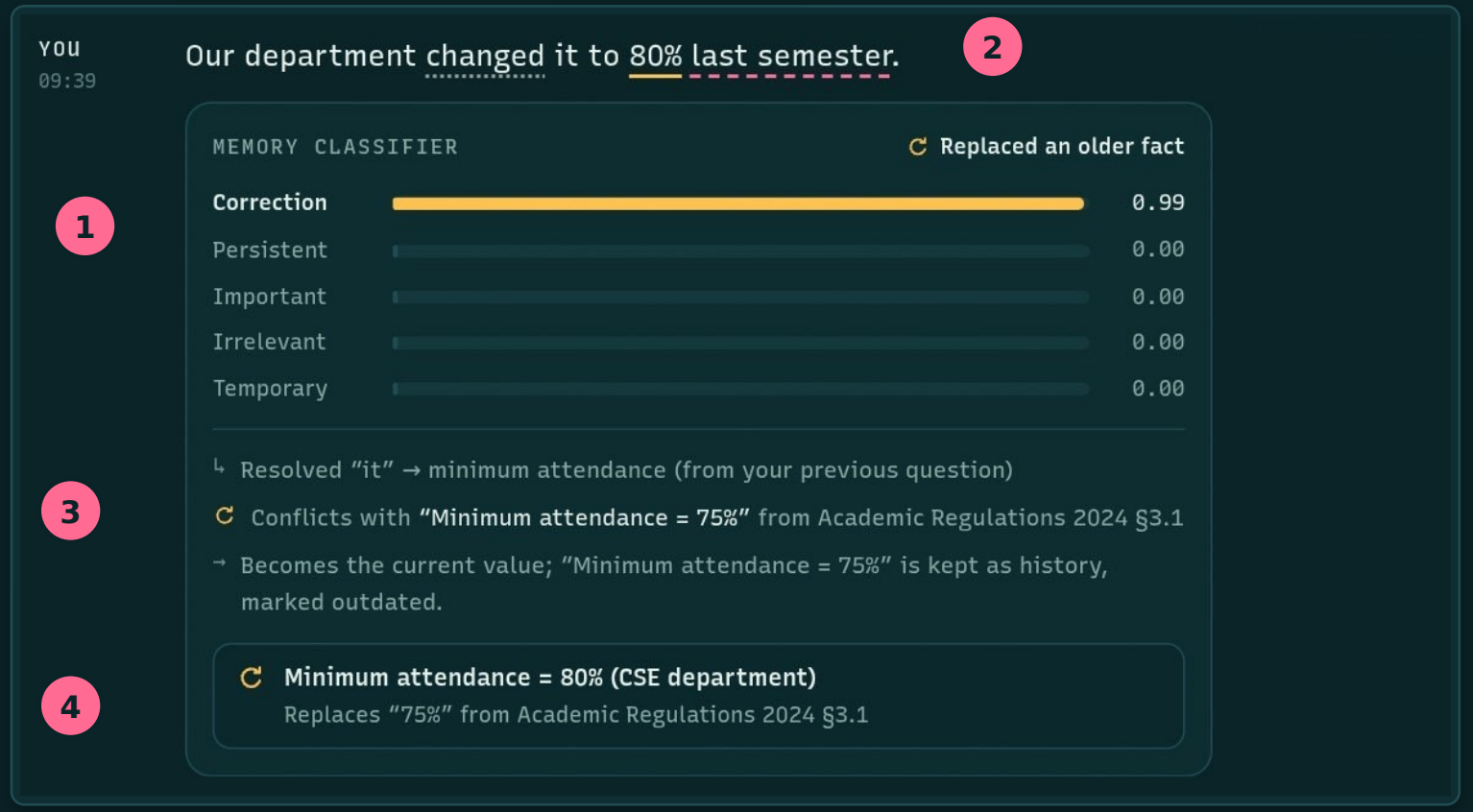
1 A question is not stored
Asking isn't telling — the classifier marks it 'Question · not stored'.

2 Answer: 75%, with citations
From Academic Regulations 2024 §3.1; every claim carries a [n] chip.

3 Inspector explains the ranking
Searched 28 passages + memories → similar enough → top 5 re-ranked.

CONSOLE · DEMO STEP 2

Correct it: the memory classifier at work



1. **Class probabilities**
   Correction 0.99 — from the trained classifier (TF-IDF + logistic regression).

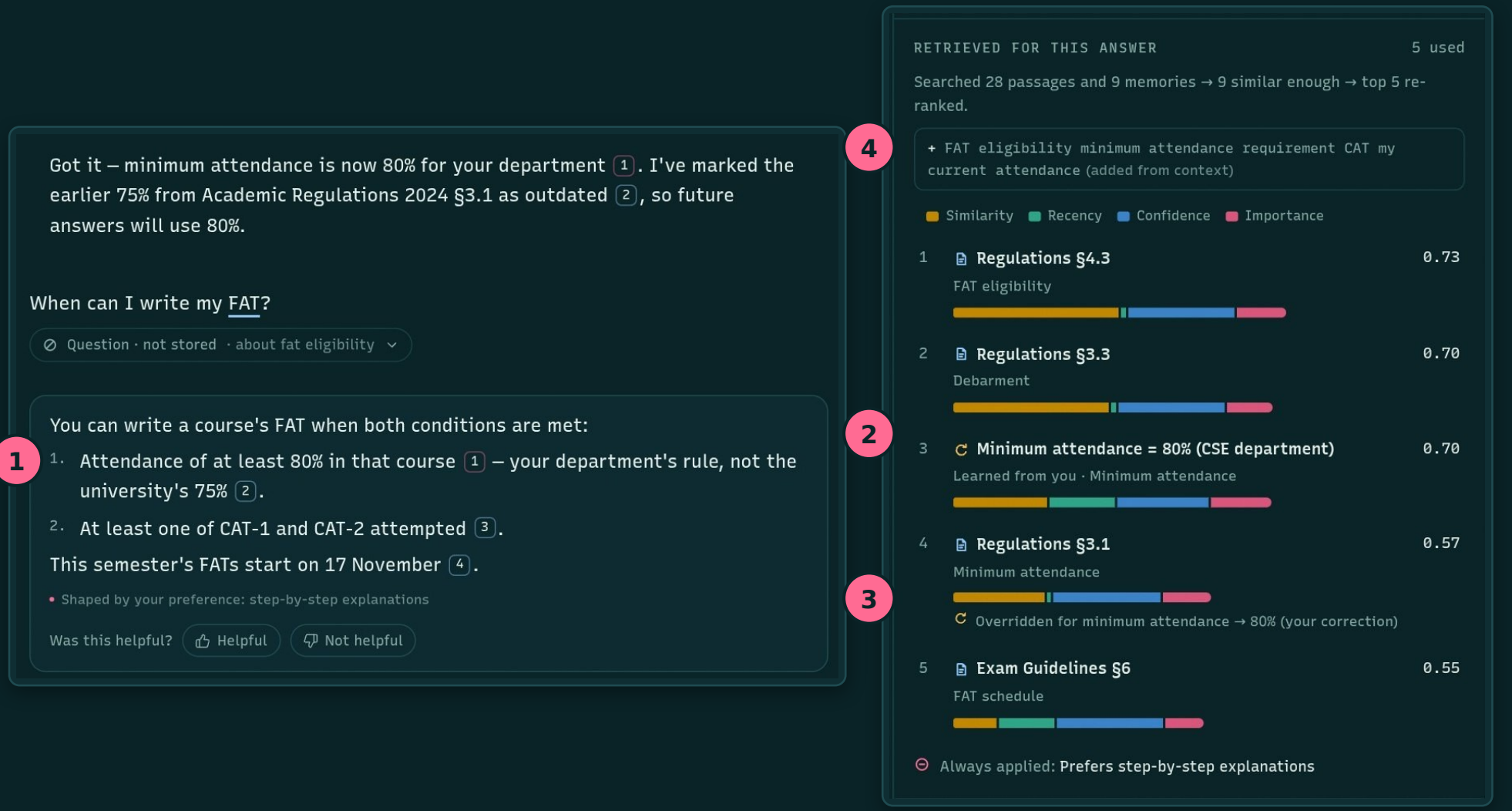
2. **Highlighted evidence**
   Cue 'changed', value '80%', time 'last semester'.

3. **Uses the conversation**
   Resolves 'it' → minimum attendance; finds the conflict with 75% in §3.1.

4. **Decision: replace**
   80% becomes current; 75% is kept as history, marked outdated.

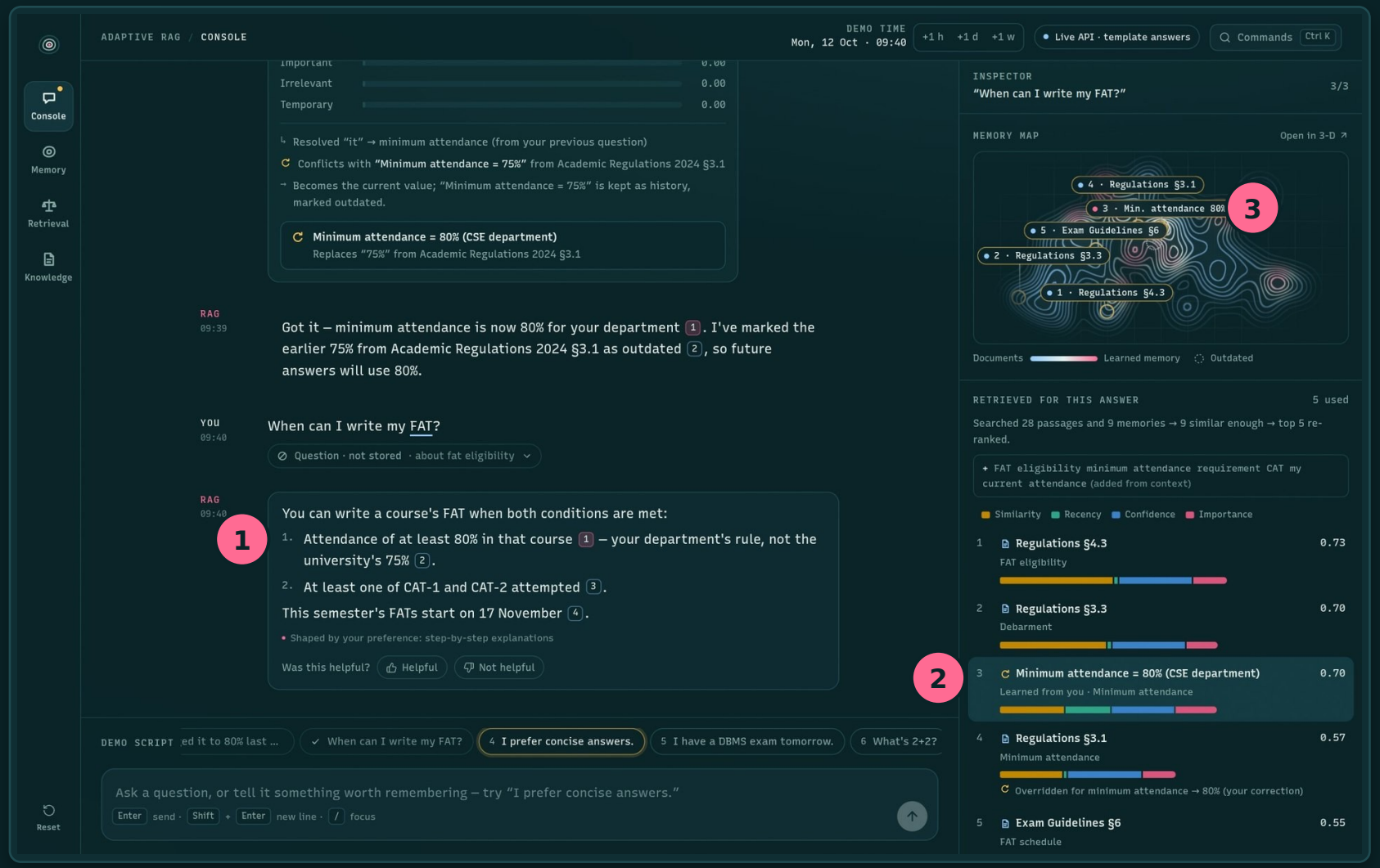
CONSOLE · DEMO STEP 3

The next answer uses the correction



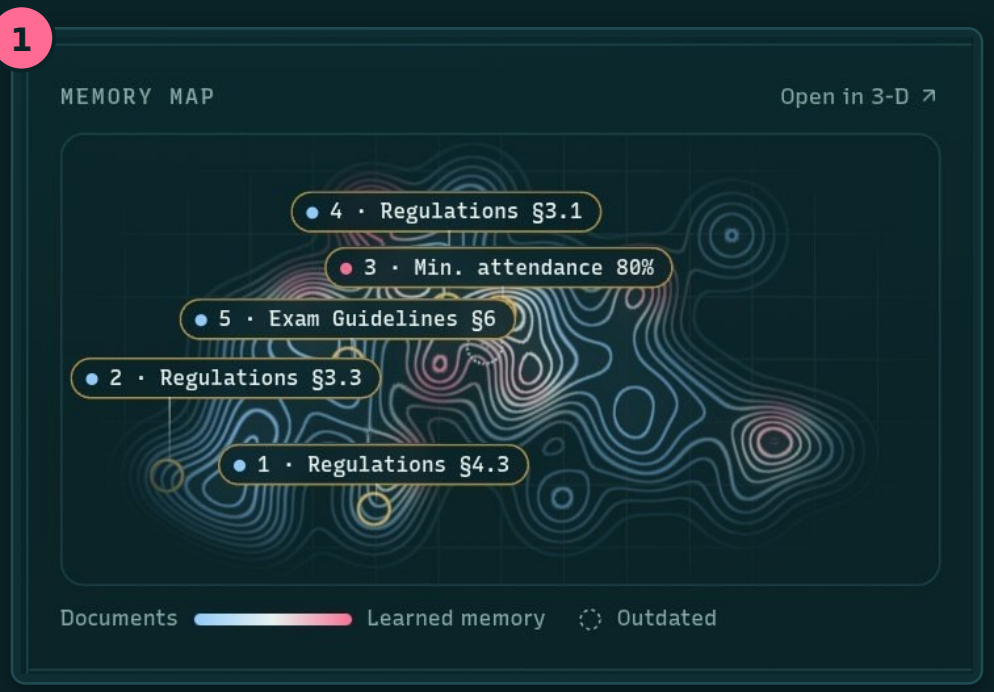
1. **Answer says 80%**
   Cites your correction and the old 75% rule.
2. **Correction ranked 3rd**
   Your memory scores 0.70, next to the regulations.
3. **§3.1 kept, overridden**
   “Overridden for minimum attendance → 80%”.
4. **Query expanded**
   Context terms added from the conversation.

Every claim is cited — and traceable



1. **Hover a citation chip**
   Chip [1] in the FAT answer.
2. **The source row lights up**
   Minimum attendance = 80% (your correction), 0.70
3. **…and its map label**
   3 · Min. attendance 80%.
4. **Click to open the source**
   A drawer shows the passage or memory, its reliability and any override.

Why each source was picked





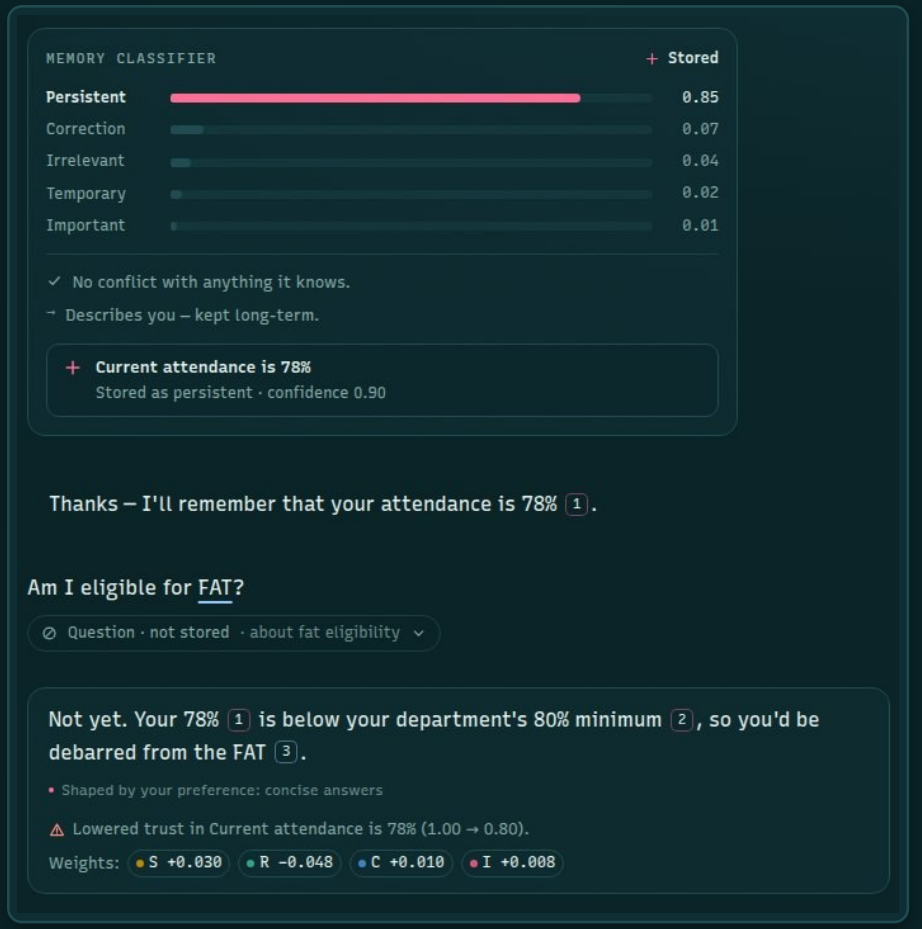


1. **Memory map**
   Ping = where the question landed; numbered rings = sources used.
2. **Ranking weights**
   The four signal weights S, R, C, I — tunable in the lab.
3. **Retrieved sources**
   Hover a score bar: how much each signal added.

CONSOLE · DEMO STEPS 4–8

Every message gets a different decision

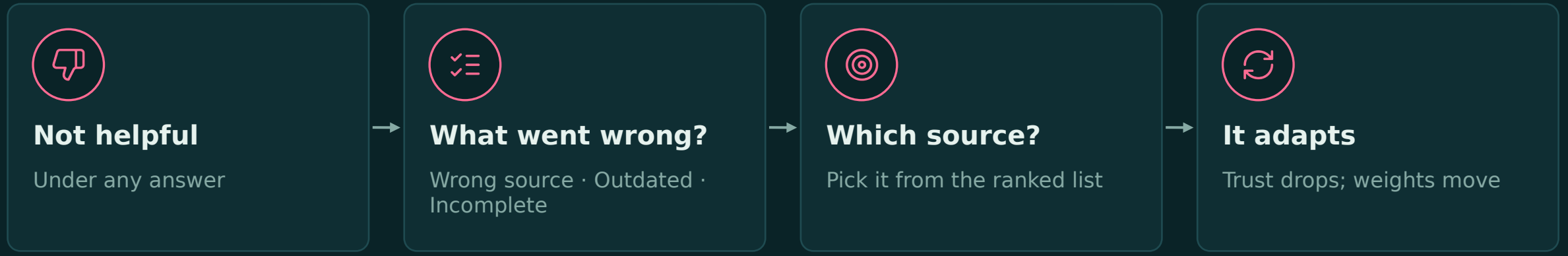
MESSAGE	CLASSIFIED AS	WHAT HAPPENS
I prefer concise answers.	Persistent · preference	Replaces the step-by-step preference; answers get shorter
I have a DBMS exam tomorrow.	Temporary	Kept until the end of tomorrow, then expires
What's 2+2?	Question · not stored	Answered directly — no documents needed
My attendance is 78%.	Persistent · about you	Stored as a fact about the student (0.85)
Am I eligible for FAT?	Question	Combines 78% + the 80% correction + debarment → “Not yet”



Right: “My attendance is 78%” stored as persistent, then the eligibility answer citing all three sources.

CONSOLE · FEEDBACK

Feedback teaches the ranking



RESULT IN THIS RUN (marked outdated)

Lowered trust in “Current attendance is 78%”

reliability 1.00 → 0.80

Weights: S +0.030 · R −0.048 · C +0.010 · I +0.008

Helpful instead → the cited sources gain trust (+0.03) and the weights lean towards what ranked them first.

THE UPDATE RULES (learning.py)

Weights (pairwise ranking update)

$w \leftarrow \text{normalise}(w + \eta \cdot \text{clip}(f_{\text{preferred}} - f_{\text{rejected}}))$

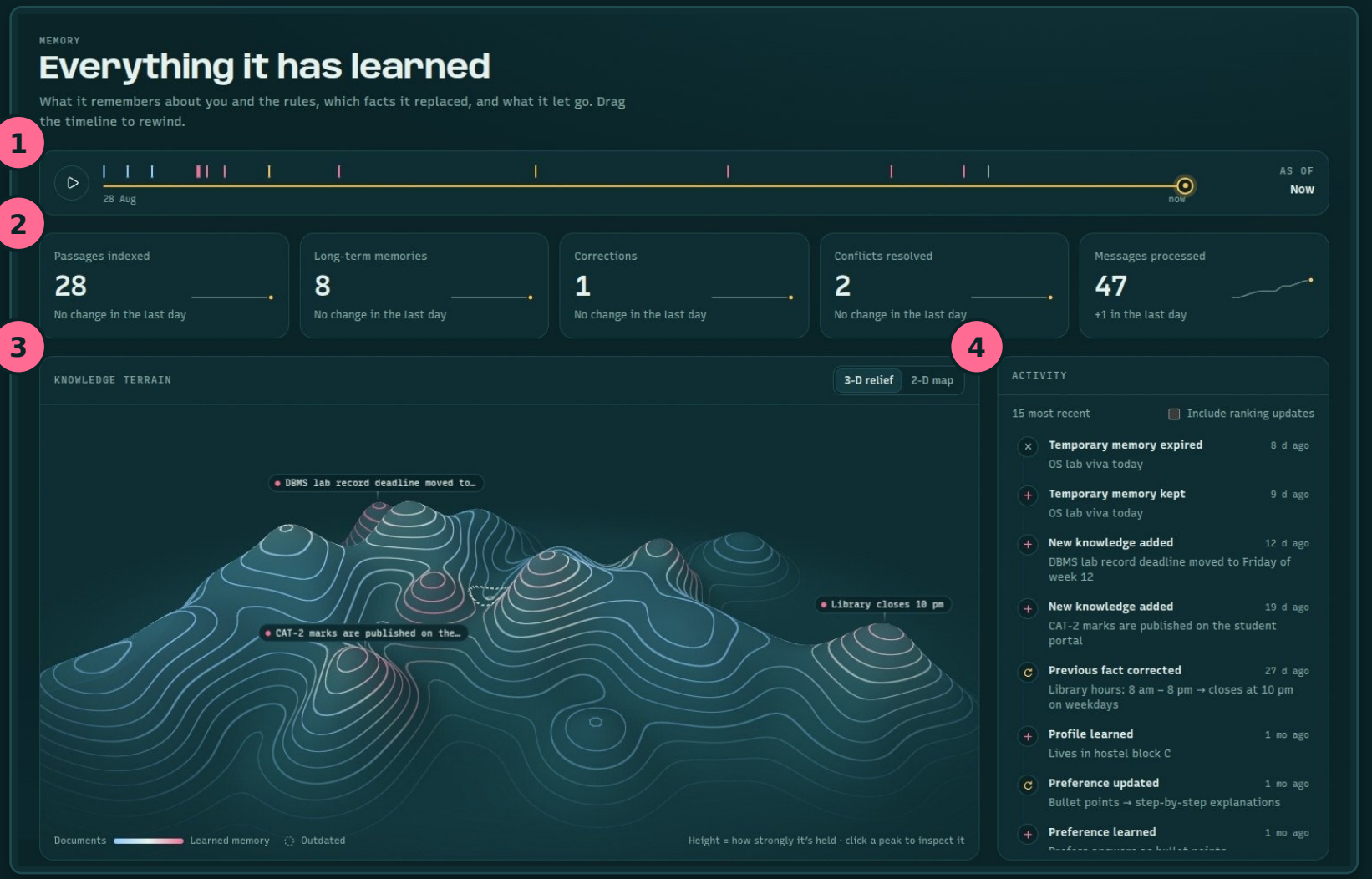
$\eta = 0.08$ · each step capped at ± 0.05 · every weight ≥ 0.02 · sum = 1

Source reliability

rejected × 0.85 (floor 0.2) · outdated × 0.8 · helpful + 0.03

MEMORY PAGE

Everything it has learned



1. **"As of" timeline**
   Drag to rewind; press play to replay how memory evolved.

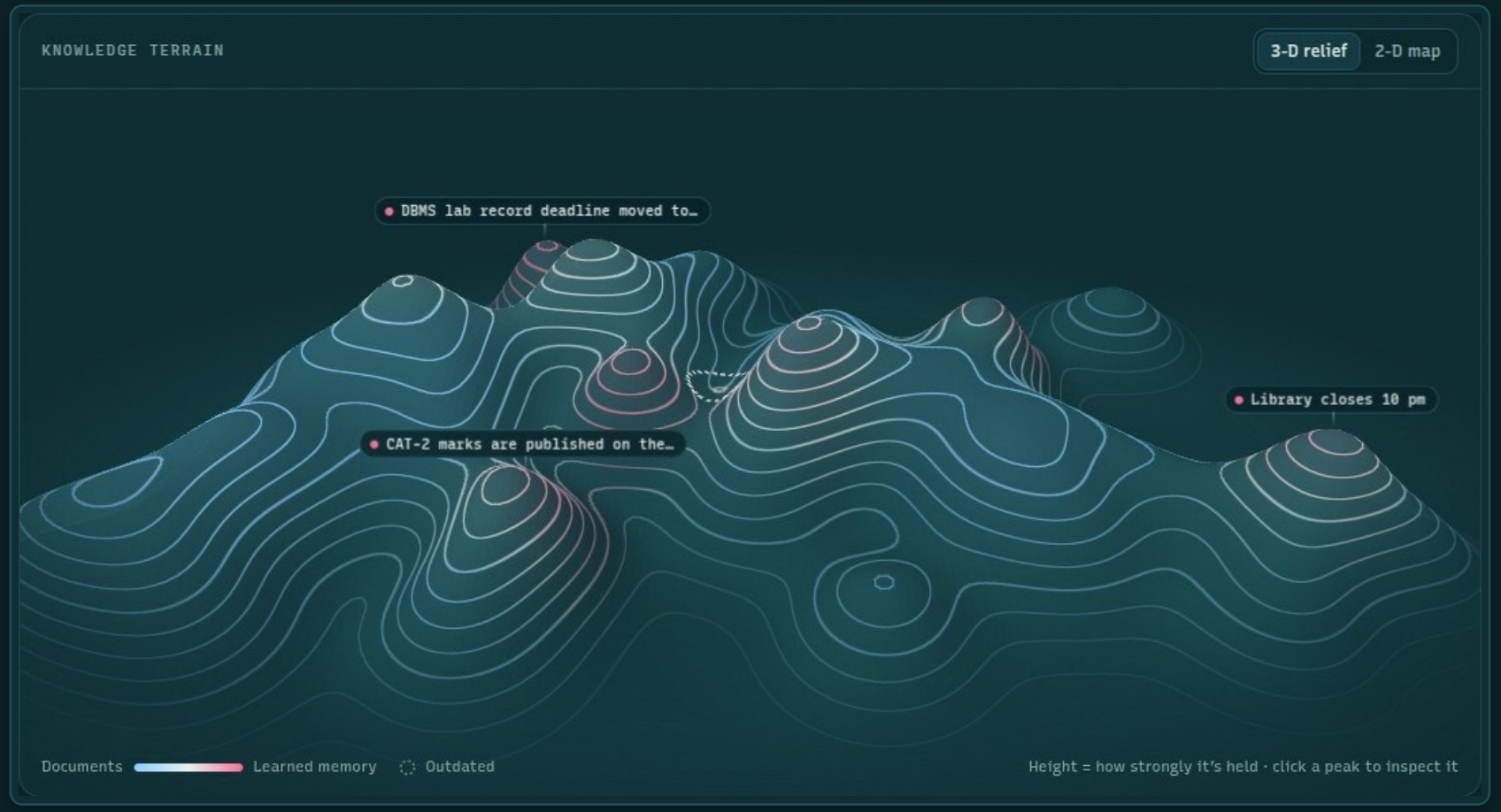
2. **Counters**
   Passages 28 · memories 8 · corrections · conflicts resolved · messages.

3. **Knowledge terrain**
   3-D relief or 2-D map; click a peak to inspect it.

4. **Activity feed**
   Every memory event: added, corrected, expired.

THE KNOWLEDGE TERRAIN

A map of everything it knows



Blue hills
Passages from the official documents

Pink peaks
Facts learned in conversation

Height
How strongly it's held — decays when unused

Distance
Meaning: similar items sit close together

Dashed ring
An outdated fact, kept as history

Appears on four pages: landing (full screen), console inspector, memory, knowledge.

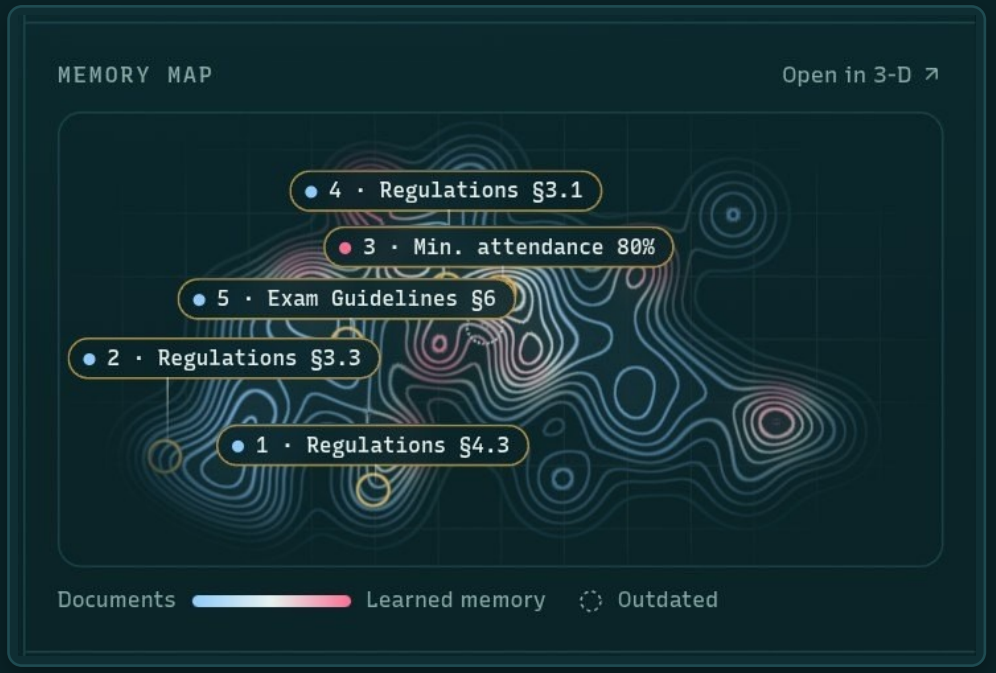
How the terrain is built

1 · Embed
Each passage and memory becomes a 384-number vector from the MiniLM sentence model.

2 · Place
Classical MDS on cosine distance → x, y. New items are placed beside their nearest neighbours.

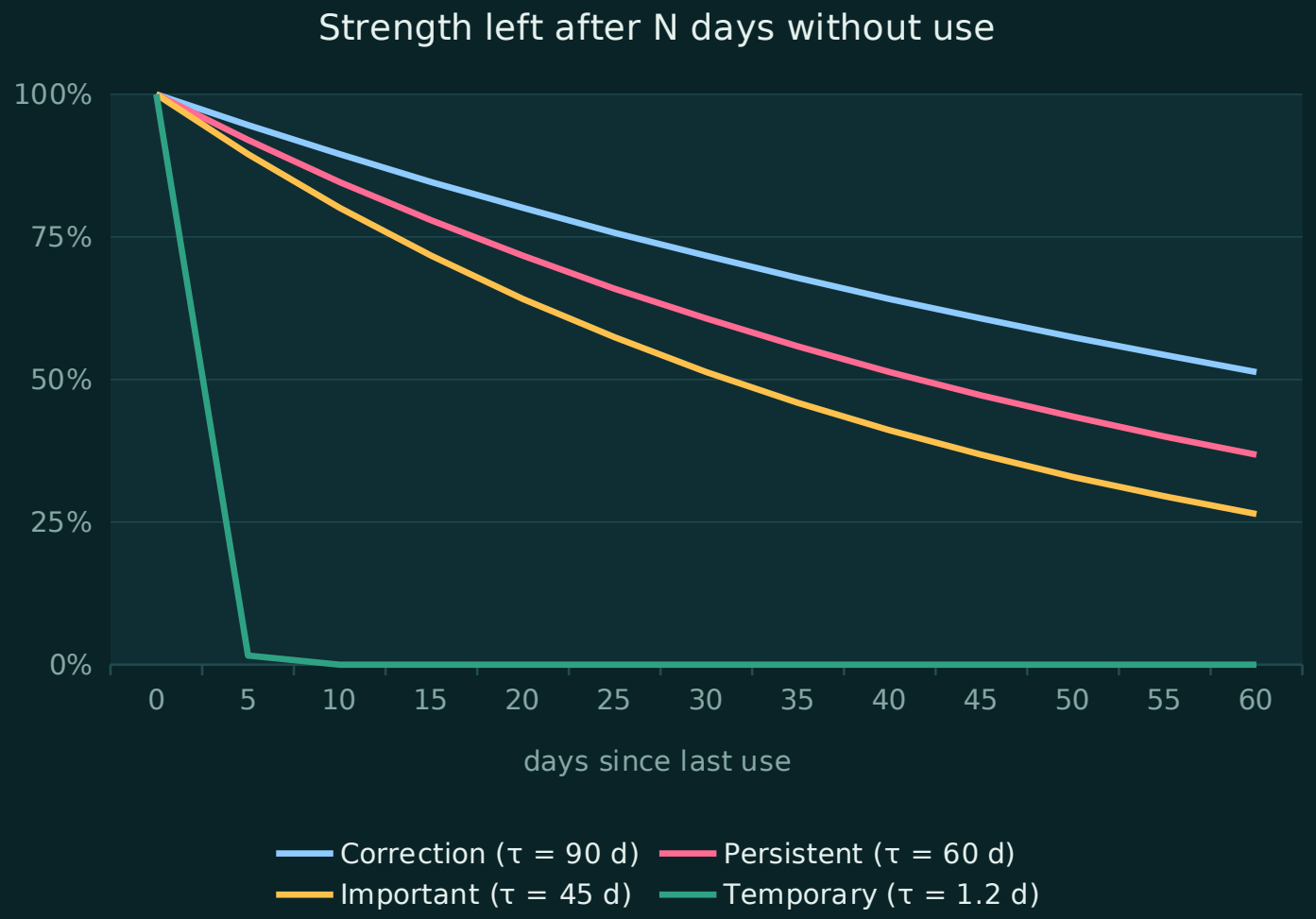
3 · Height
Memory: 0.22 + 0.85 × strength
Passage: (0.34 + 0.3·importance) × (0.55 + 0.45·reliability)

4 · Render on the GPU
Sum of Gaussian peaks on a 260×150 grid, normalised; 14 contour levels; colour blue → pink by memory share.



Live signals on the map
Amber ping = where a question lands. Numbered rings = the sources used. Ripples = a memory was added, replaced, flagged or expired.

Memories fade unless they're used



strength = importance × confidence × e^(−Δt/τ)
τ = base(class) × 1.2^uses
Each use stretches τ by 20% (up to 8 uses).

1. **Corrections last longest**
   τ = 90 days; persistent 60; important 45.

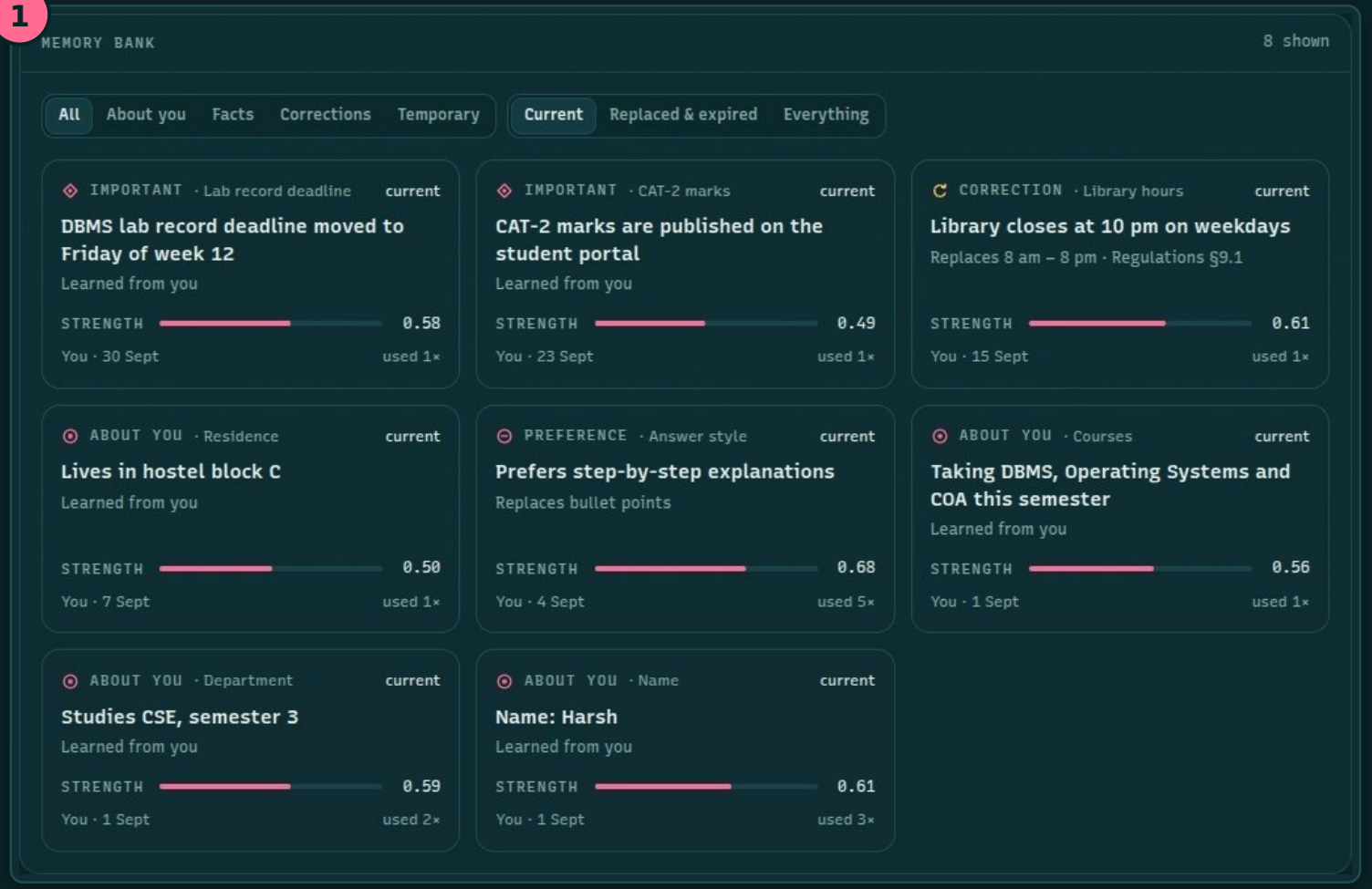
2. **Temporary facts expire**
   “DBMS exam tomorrow” is kept until the end of tomorrow.

3. **Time travel to test it**
   +1 h / +1 d / +1 w or demo step 9 jumps the clock past the exam.

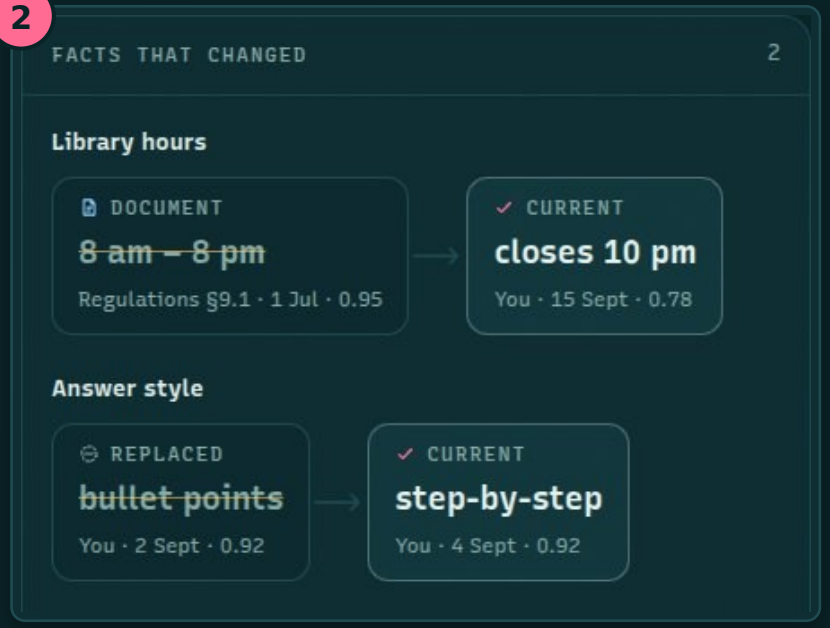
4. **Pin to keep forever**
   Pinned memories don't decay.

Memory bank and facts that changed

1



2



1 **Cards with strength bars**
Filter by type and status (Current · Replaced & expired · Everything).

2 **Version chains**
Old value struck through → current value, with source, date and confidence.

Rewind memory to any moment

1



“As of” timeline — every tick is a memory event; drag to rewind, ▷ to replay

2

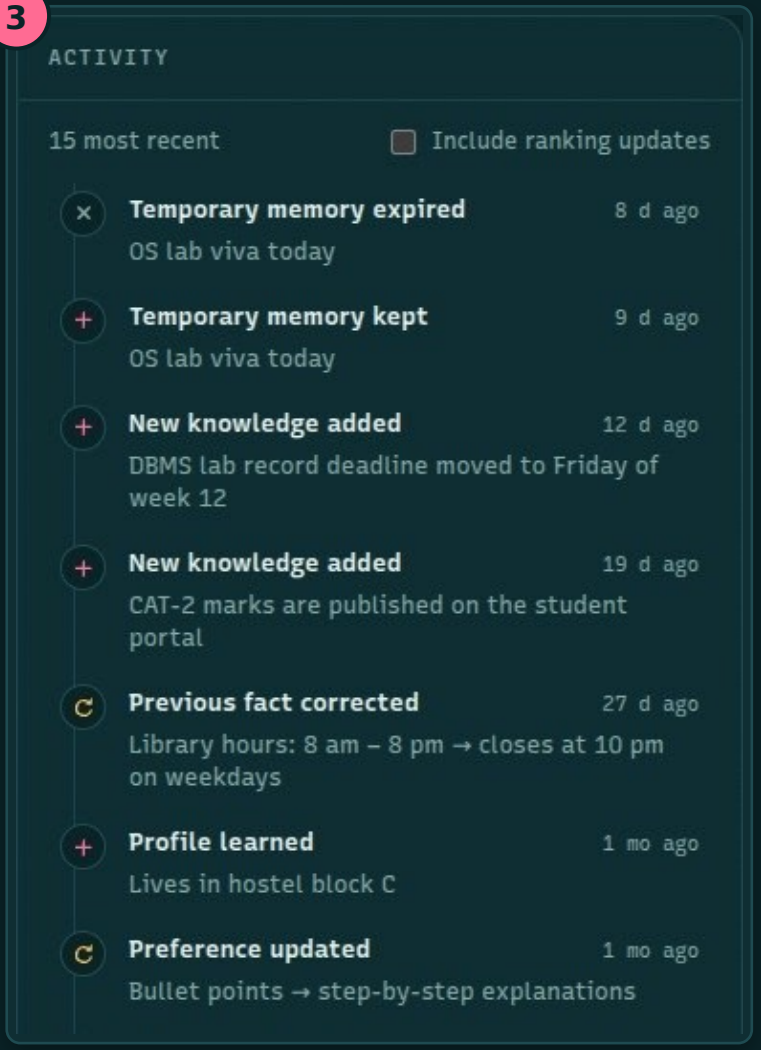


Counters with 12-day sparklines and day-over-day change

3 **Activity feed**
Every memory event, newest first: kept, expired, added, corrected, updated.

Rewinding recomputes everything. Terrain heights, card status and counters are recalculated for that moment. **Replay:** press ▷ to watch memory evolve from the first day to now.

3



RETRIEVAL LAB

The ranking formula, four signals

$$\text{score} = w_1 \cdot S + w_2 \cdot R + w_3 \cdot C + w_4 \cdot I$$

S — weight now 0.45

Similarity
Calibrated cosine similarity between the question and the item's embedding.

R — weight now 0.15

Recency
e^(−age/τ): τ = 30 days for memories, 365 for documents.

C — weight now 0.25

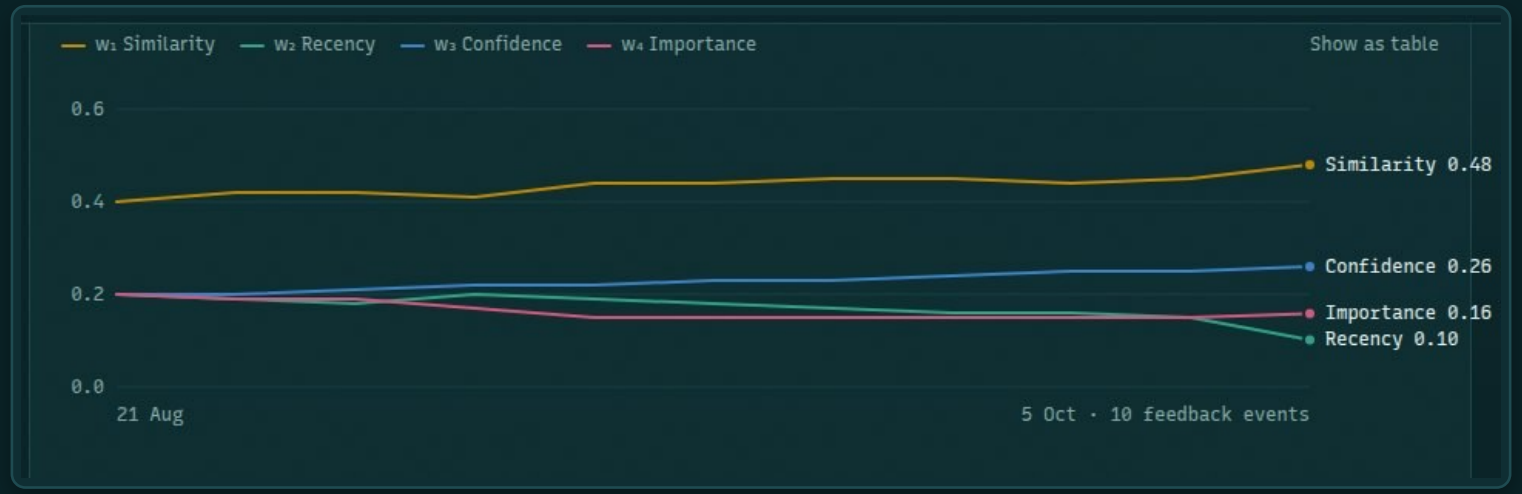
Confidence
Confidence × reliability — the part feedback changes.

I — weight now 0.15

Importance
From the memory class or the document section.

Two-stage retrieval: vector search keeps the 12 nearest items (similarity ≥ 0.2), the formula re-ranks them, the top 5 go to the model. In the lab, “Try your own” lets you drag the weights (they always sum to 1) and watch rows move ▲▼ before applying them to the assistant.

What feedback changes



How feedback moved the weights — 10 feedback events since 21 Aug

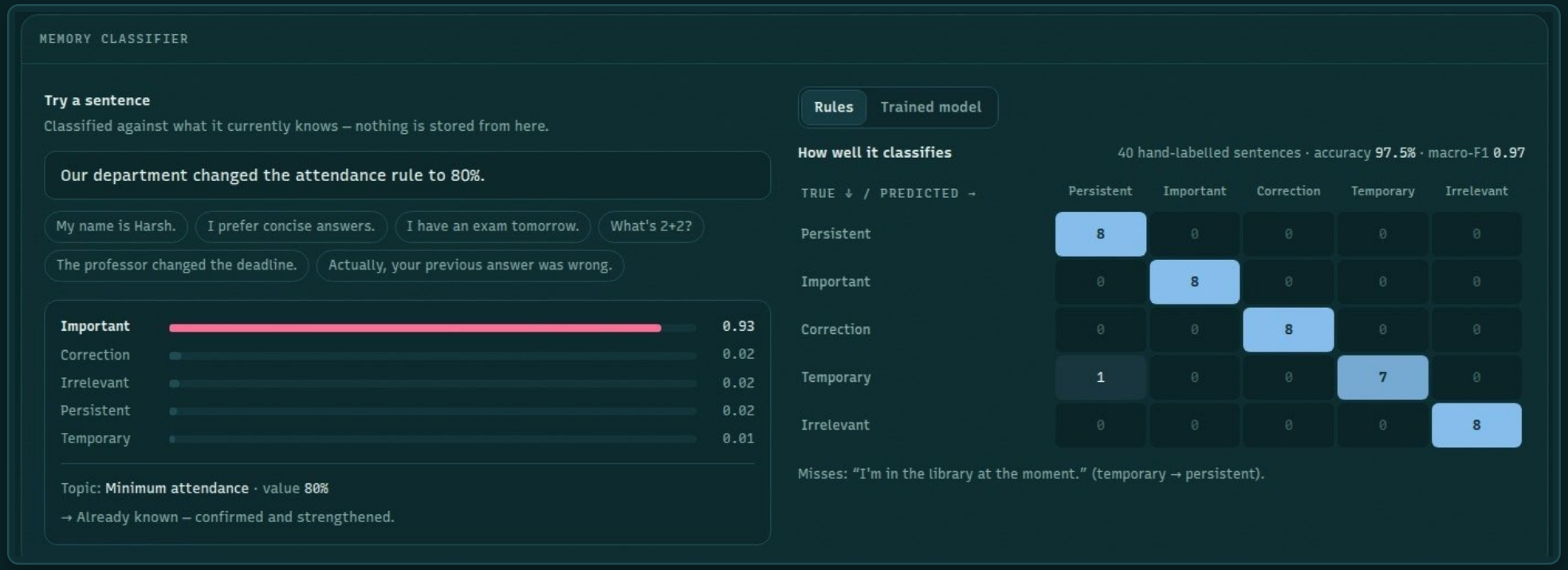
1. **Similarity ↑ 0.40 → 0.48**
   Feedback rewarded sources that matched the question closely.

2. **Recency ↓ 0.20 → 0.10**
   Newer wasn't always better for these questions.



Source reliability: trust lowered by feedback multiplies a source's confidence (C).

The classifier, tested on sentences it never saw



97.5%

Rules baseline
39 of 40 held-out sentences · macro-F1 0.97

100%

Trained model
40 of 40 · TF-IDF + logistic regression

0.88

CV macro-F1
5-fold cross-validation on 220 training sentences

KNOWLEDGE PAGE

The documents it answers from

STAGE 1

Add

.txt · .md · .pdf, or the sample CSE circular

STAGE 2

Split

Sections → passages of ~110 words; claims extracted

STAGE 3

Embed

Each passage → a 384-d vector

STAGE 4

Index

Stored in the Chroma vector store

STAGE 5

Reconcile

Confirm a memory, or resolve a conflict

Documents

Academic Regulations 2024 (14 passages) · Exam Guidelines (6) · Hostel Handbook (5) · CSE Handbook (3)

Passages

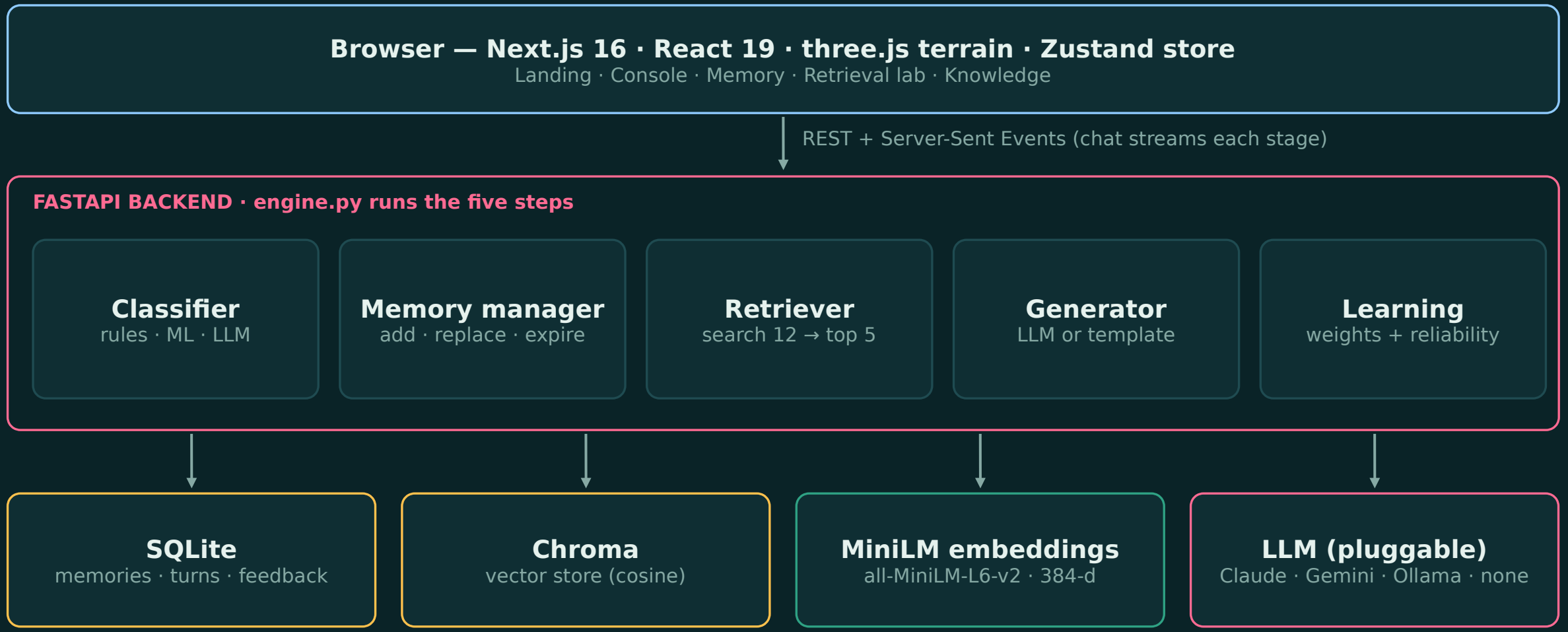
Section, title, tokens, reliability meter, embedding fingerprint strip, and “Overridden for … by …” notes

Where it sits on the map

The terrain rings the selected document's passages; new hills rise when a file is added

ARCHITECTURE

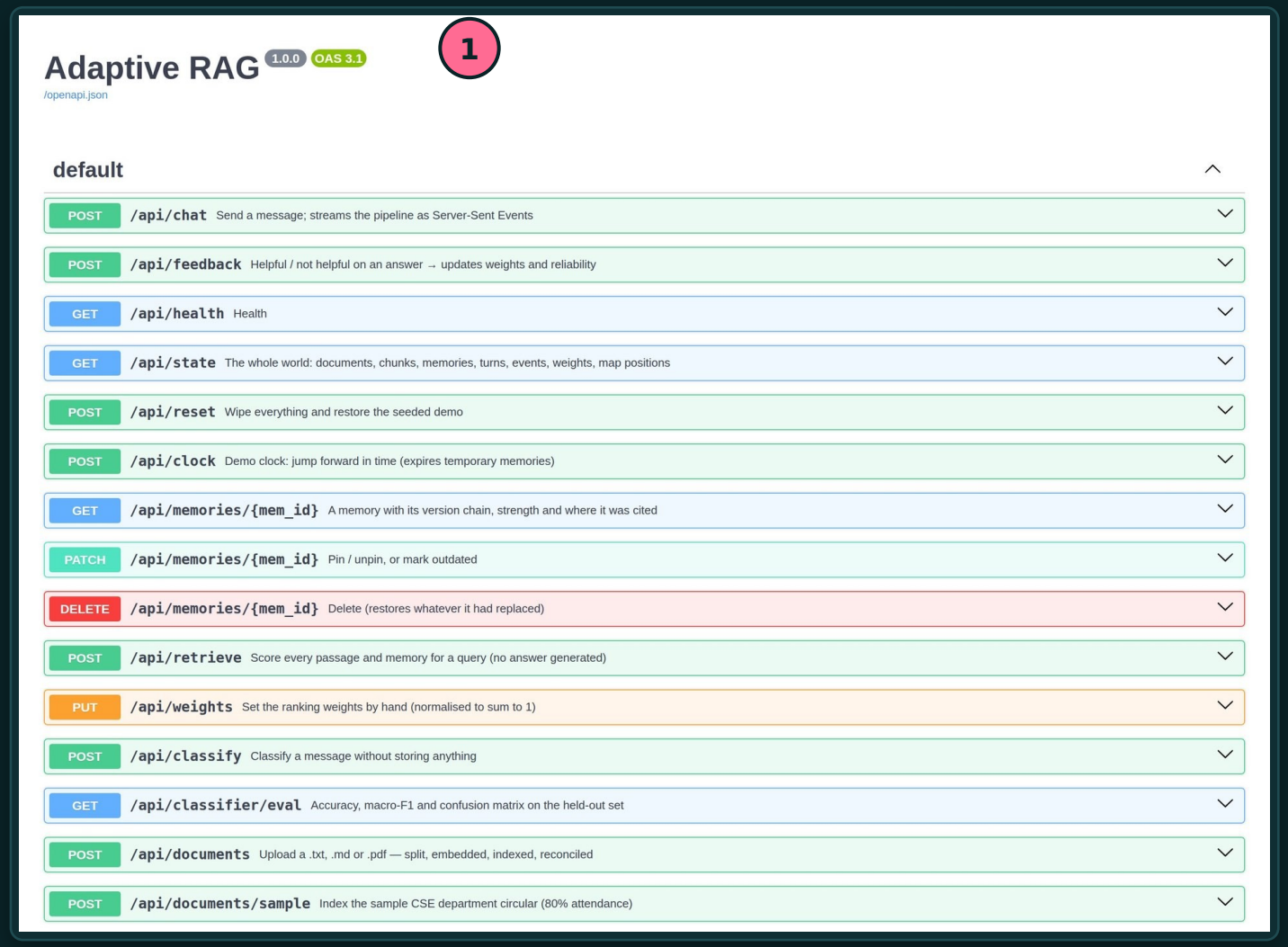
How the pieces fit together



Storage and models — the LLM is pluggable (Claude · Gemini · Ollama · none) and is never trained.

BACKEND API

15 endpoints, documented automatically



POST /api/chat STREAMS, IN ORDER

2



1 **Interactive docs at /docs**
Generated from the route definitions in main.py.

2 **One streamed request per turn**
The browser renders each event as it arrives.

CONNECTING A LANGUAGE MODEL

The LLM writes the answer — it is never trained

Provider	Set in backend/.env	Notes
Claude	ANTHROPIC_API_KEY	claude-opus-5-5 · official anthropic SDK · cached system prompt · refusal fallbacks
Gemini	LLM_PROVIDER=openai OPENAI_BASE_URL=…/v1beta/openai/	gemini-3.8-flash · tested live · free tier ≈ 20 requests / model / day
Ollama (local)	OPENAI_BASE_URL=http://localhost:11434/v1	Any local model, free, offline
None	—	Template answers from the same retrieved context — the demo still works



What the model sees: a fixed system prompt, the last few turns, the student's facts and style, and a numbered CONTEXT block — each item with its source, trust and any override. It must cite [n] and use only that context. If it fails, the turn falls back to the template answer.

TESTING

Tested end to end

44

backend tests pass

pytest · memory manager, API routes, LLM path

1 = 1

parity test

Python and TypeScript engines agree on a 14-message script

25 / 25

end-to-end checks

Headless browser runs the 9-step demo · 0 console errors

40 / 40

held-out sentences

Trained classifier (CV macro-F1 0.88)

HONEST LIMITS

Limitations and next steps

Limitations

- Classifier tested on only 40 held-out sentences; training data is synthetic
- One user, no login or per-user memory
- Free Gemini tier: about 20 live answers per model per day
- Trust only changes through feedback — no automatic fact-checking

Next steps

- Evaluate on real student messages
- Multiple users with authentication and private memory
- Combine the trained and LLM classifiers (voting)
- Measure answer quality end to end, not just ranking

THANK YOU

Correct it once. It remembers.

- Corrections replace old facts — they don't pile up.
- Retrieval ranks on four signals the system learns from feedback.
- The language model never changes — everything around it learns.

Questions?

